QAOAs and Geometry: A Capacitated Vehicle Routing Problem (CVRP).

QQueen: Catherine Kirk, Larik Lu, Ivan J. Paiewonsky

Formulation

Input: one depot, customer locations, number of vehicles N, vehicle capacity C

Decision: assign each customer to a vehicle and determine the visiting order

Objective: minimize the total travel distance

Constraints:

-each route starts and ends at the depot

-each customer is visited exactly once

-each vehicle serves at most C customers

-only has N vehicles

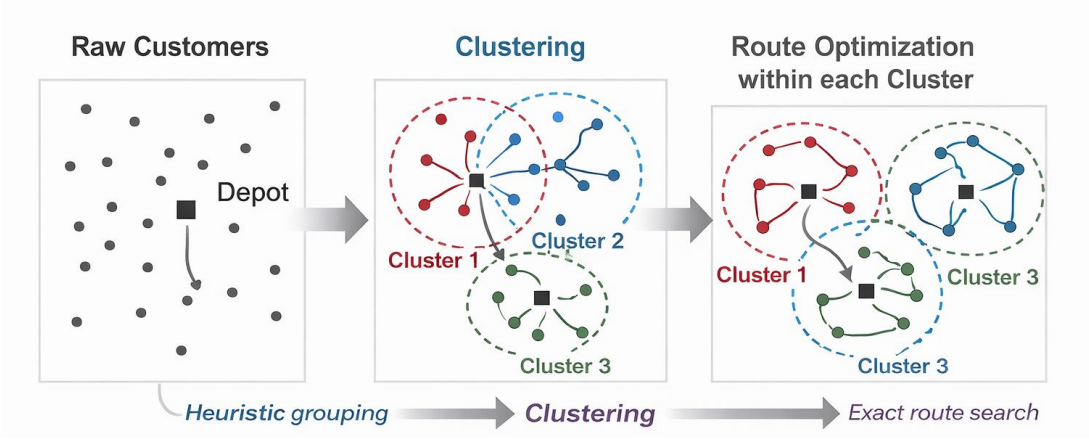
$$\boxed{\min \sum_{\text{all vehicle routes}} \text{travel distance}}$$

Project Logic

1. Build a classical model for the CVRP
2. Replace components with a quantum-classical hybrid module
3. Compare the two approaches in terms of its computational behavior and accuracy
4. Evaluate whether hybridization can improve classical computing

Classical Model

Model Structure



-**Overall**: cluster-first, route-second

-**Global level**: heuristic grouping of customers -> determine clusters

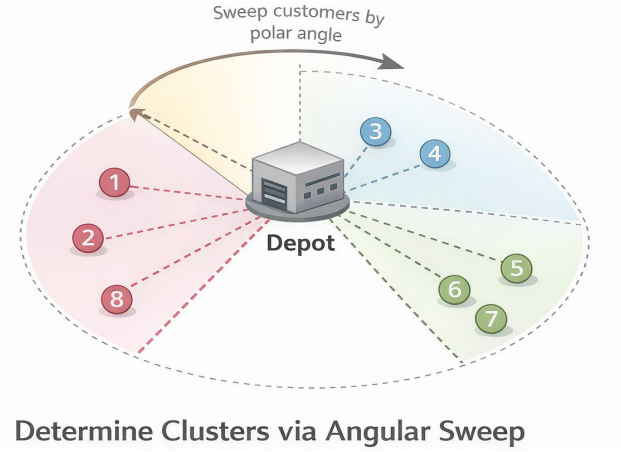
-**Within each cluster:** exact brute force search for the best route order

-Therefore: globally heuristic but locally exact

Model Pipeline

1. **Sort** customers by their polar angles
2. Estimate the number of clusters
3. Determine the customer **grouping**
4. Solve the **best route** within each cluster
5. Compute the total distance and output the solution

Determine customer grouping



- Use **angular sweep clustering** based on customer polar angles
- Estimate # of groups

$$G = \min(\lceil H/C \rceil, N)$$

H: # of customers. C: vehicle capacity. N is # of vehicles. G: # of clusters.

- Generate possible groupings through **cyclic shifts** of the ordered customer list
- For each possible grouping, **estimate the route** using a nearest-neighbor distance estimate
- Select the grouping with the smallest estimated total distance

Best Routing Within Each Cluster

- After fixing the grouping => solve a routing problem within each cluster
- **Enumerate** all customer permutations in that cluster
- Compute the route length for each permutation: depot->permutation->depot
- Select the route with the minimum distance

exact brute-force search

Classical Baseline Results

1. Customer grouping (clusters)
2. Vehicle routes
3. Total travel distance

Example result

Instance 1 | N=2, C=5

G = 1 vehicle

Groups:

G1: customers [3, 2, 1]

Routes:

r1: 0 -> 3 -> 2 -> 1 -> 0

Total distance: 21.7445

Limitations

- The grouping step is restricted by **polar-angle proximity**, so it may miss the global optimal partition
- Grouping is determined using a **distance estimate**, not the exact CVRP
- Overall solver is therefore **globally heuristic**, even though routing within each cluster is exact

Future steps

- Replace the heuristic grouping step with a more global optimization method
- Test the approach on larger CVRP instances

Hybrid Model:

QAOA for Cut-Point Selection

Mathematical Reasoning

After sorting, there exists $H-1$ gaps between consecutive customers. $G-1$ cuts song these gaps defines a partition into G groups. Each cut represents a route through the depot instead of between neighboorhoods.

This reduces the exponential grouping problem to a binary optimization over H-1 variables: $b_i = 1$ if we cut at gap $i, b_i = 0$ otherwise. The detour cost of cutting at gap i is

$$w_i = d(c_i, \text{depot}) + d(\text{depot}, c_{i+1}) - d(c_i, c_{i+1})$$

Low w_i means the two customers are far apart relative to the depot, so cutting there is cheap.

QUBO (**Quadratic Unconstrained Binary Optimization**)

$\longrightarrow$ Any constrained binary optimization can be converted to QUBO by moving constraints into penalty terms, which is what we do with the cut-count constraint.

$\longrightarrow$ The substitution $b_i = (1 - Z_i)/2$ converts any QUBO into a sum of Pauli-Z operators, which is exactly the form QAOA requires for its cost unitary

$\longrightarrow$ Ising Hamiltonians composed of Z and ZZ terms are diagonal in the computational basis, meaning the cost unitary $e^{(-i\gamma H_C)}$ decomposes into single-qubit R_Z gates and two-qubit R_{ZZ} gates with no additional entangling overhead.

QUBO

In this case, QUBO optimization formulas map directly to an Ising Hamiltonian via a simple variable substitution.

The objective function is

$$C(b) = \sum_i w_i b_i + \lambda \left(\sum_i b_i - (G-1) \right)^2$$

The first term minimizes total detour cost. The second term is a penalty that enforces exactly $G-1$ cuts. λ is set to $2n \max(|w_i|)$ to ensure the penalty dominates any single detour cost, making invalid solutions energetically unfavorable.

Ising Hamiltonians

$$H_C = \sum_i [-(w_i/2) - \lambda A] Z_i + (\lambda/2) \sum_{i<j} Z_i Z_j$$

Where $A = n/2 - (G-1)$ and $n = H - 1$ is the number of qubits.

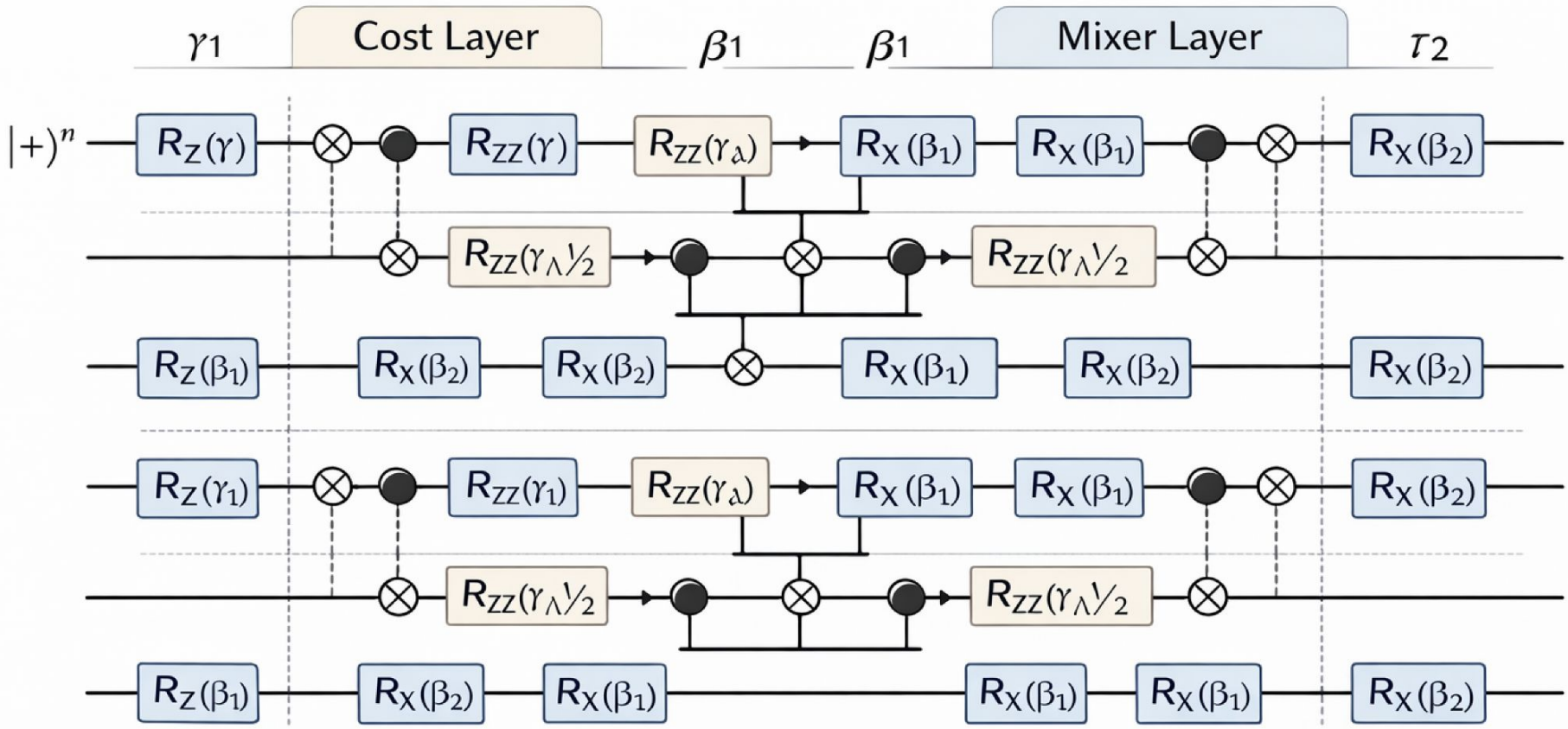
The linear Z_i terms encode the individual detour costs adjusted by the penalty. The ZZ coupling terms arise from the quadratic penalty and connect every pair of qubits (all-to-all connectivity), reflecting the global nature of the cut-count constraint.

Circuit Structure

The QAOA ansatz alternates two unitaries for p rounds (we use $p = 2$):

- **Initial state:** $|+\rangle^n$ (uniform superposition via Hadamard on each qubit).
- **Cost layer:** $\exp(-i\gamma_k H_C)$. Since H_C is diagonal in Z, this decomposes into $R_Z(\gamma \cdot \text{coeff})$ gates on each qubit (from the linear terms) and $R_{ZZ}(\gamma \cdot \lambda/2)$ gates on each qubit pair (from the ZZ terms). The R_{ZZ} gates are implemented via two CNOT gates sandwiching an R_Z rotation.
- **Mixer layer:** $\exp(-i\beta_k H_M)$ where $H_M = \sum_i X_i$. This decomposes into independent $R_X(2\beta)$ gates on each qubit.

The circuit has $2p = 4$ variational parameters: $\{\gamma_1, \beta_1, \gamma_2, \beta_2\}$.


γ1
Cost Layer
β1
β1
Mixer Layer
τ2
|+)^n
RZ(γ)
RZZ(γ)
RZZ(γ1)
RX(β1)
RX(β2)
RZ(β1)
RZ(γ1)

COBYLA: (Constrained Optimization By Linear Approximations) is a gradient-free optimizer

- The QAOA objective landscape is noisy, non-convex, and riddled with local minima. Gradient-based methods (e.g., L-BFGS-B, Adam) require smooth gradients.
- COBYLA uses linear approximations of the objective within a trust region, making it robust to noise.
- COBYLA has low per-iteration cost: it requires only function evaluations (no gradient circuits), which matters when each evaluation involves a statevector simulation. We cap at 200 iterations, which is sufficient for 4 parameters.

Our design

Step 1 (Classical): Encode customer locations, compute the Euclidean distance matrix, sort customers by polar angle from depot.

Step 2 (Quantum): Build cut-point weights w_i, construct the cost Hamiltonian as a SparsePauliOp, instantiate QAOA Ansatz with reps=2, optimize parameters via COBYLA using exact statevector expectation values.

Step 3 (Sampling): Transpile the optimized circuit for the target backend (IBM hardware for up to 15 qubits, AerSimulator beyond that). Sample 2048 shots. Extract the highest-probability bitstring with exactly G-1 ones.

Step 4 (Classical): Partition customers according to the QAOA-selected cuts. Find optimal intra-group routes via exhaustive permutation. Output solution files.

Scalability

- Qubit count scales as $H-1$ (one qubit per angular gap). For the instances tested: $0, 2, 5, 11$ qubits.
- Gate count scales as $\mathcal{O}(n^2)$ due to all-to-all ZZ couplings in the penalty term. Each pair of qubits requires an R_{ZZ} gate (implemented as 2 CNOTs $+1$ R_Z), so the two-qubit gate count grows quadratically.
- Circuit depth also grows quadratically in the worst case, though transpiler optimization can reduce this depending on the backend's native connectivity.
- COBYLA iterations are bounded at 200, but convergence typically occurs in 35 to 50 iterations for these problem sizes. Each iteration requires one statevector simulation, which is $\mathcal{O}(2^n)$ in memory and time.

Results

Instance	Qubits	Gates	Depth	Backend	Time (s)
1	0	0	0	none (G=1)	0.00
2	2	26	18	ibm_fez	0.06
3	5	172	109	ibm_fez	0.03
4	11	1040	430	ibm_fez	0.16

Limitations

The QUBO enforces exactly G-1 cuts but does not constrain the maximum number of customers per group. This leads to capacity violations in Instances 3 and 4

the classical optimization loop uses exact statevector simulation, which is exponential in qubit count. Beyond approximately 25 to 30 qubits, this becomes infeasible on classical hardware, requiring a shift to shot-based optimization

the solver only considers cuts along the angular sweep. It cannot discover groupings that mix customers from non-adjacent angular positions, which may be optimal for some spatial configurations.

1. Classical problem | 2. Classical solution | 3. Quantum formulation | 4. Hybrid solution

Step 0/10

Total distance: 59.54